PHASE REPORT / PERSONAL TECHNICAL ARTIFACT

PACTA-BASELINE-STABILIZATION

A compact artifact of work completed, decisions made, and follow-on paths worth exploring.

project :: PACTA Vietnam

date :: 2026-04-16

repo :: pacta-vietnam

PROJECT

PACTA Vietnam

REPOSITORY

pacta-vietnam

DATE

2026-04-16

01 / INPUT -> OUTPUT

Input -> Output Summary

INPUTS

- Synthetic Vietnam loanbook, ABCD, market-share, and CO2 scenario files already present in `data/`
- Existing Vietnam pipeline in `scripts/pacta_vietnam_scenario.R`
- Known blocker context from `scripts/debug_ms.R` and prior progress notes

OUTPUTS

- Completed Vietnam PACTA run with 43 loans matched at 100% coverage
- Rebuilt SDA outputs, 2030 alignment tables, and refreshed charts in `synthesis_output/vietnam/`

- Confirmed stakeholder report in `reports/PACTA_Vietnam_Bank_Report.html`

02 / LOGIC FLOW

Logic Flow

03 / MATH / ALGORITHM USED

Math / Algorithm Used

- Fuzzy matching uses Jaro-Winkler with `min_score = 0.8` after ASCII normalization for Vietnamese names.
- Market-share alignment compares projected 2030 technology share against `target_pdp8_ndc` and treats low-carbon technologies as aligned only when they meet or exceed target.
- SDA alignment compares projected 2030 emission intensity against the synthetic PDP8/NDC target and marks any value above target as misaligned.

04 / TOOLS / FORMULAS / METHODS

Tools / Formulas / Methods

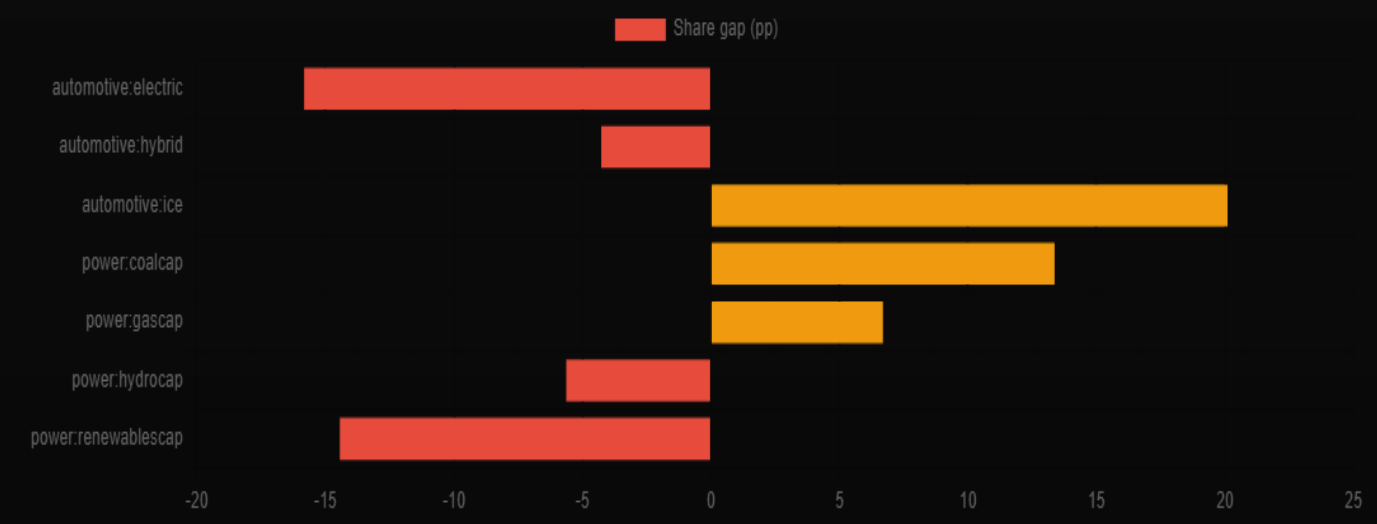
ITEM	DETAIL
Pipeline	scripts/pacta_vietnam_scenario.R
Debug helper	scripts/debug_ms.R

ITEM	DETAIL
Key R packages	pacta.loanbook, r2dii.match, r2dii.analysis, r2dii.plot
Primary outputs	06_vn_ms_alignment_2030.csv, 06_vn_sda_alignment_2030.csv, PACTA_Vietnam_Bank_Report.html

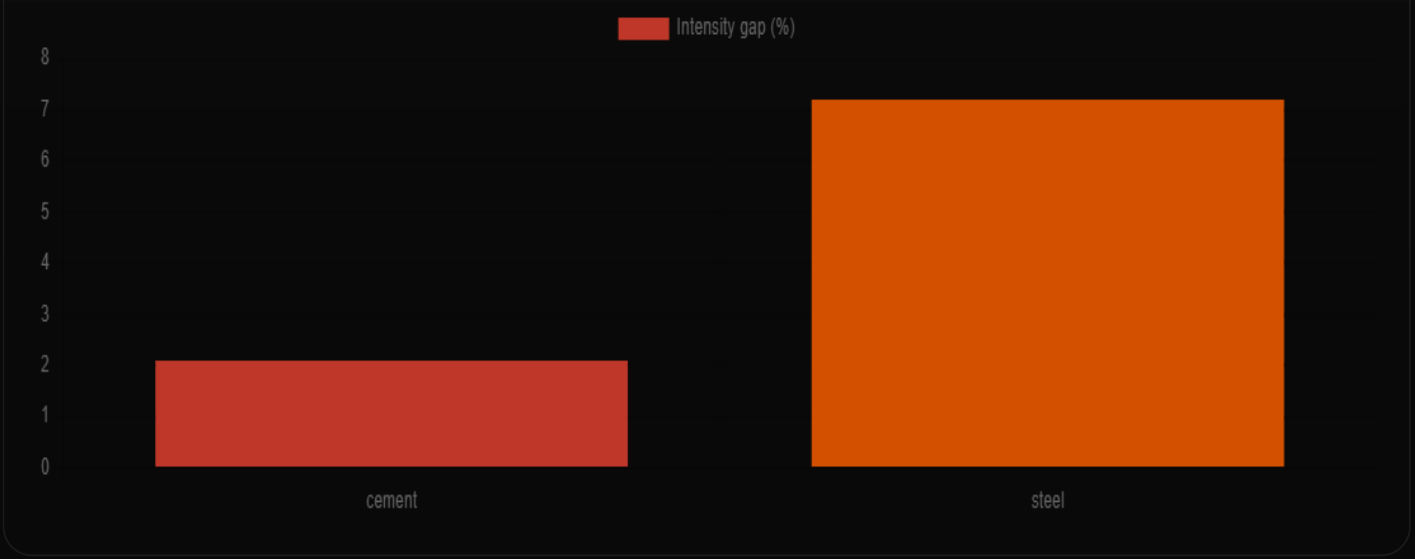
05 / RESULTS & VISUALIZATIONS

Results & Visualizations

2030 Market-Share Gaps vs PDP8/NDC



2030 SDA Gap vs PDP8/NDC



06 / LIMITATIONS / SECOND-BEST ALTERNATIVE

Limitations / Second-Best Alternative

- The pipeline now completes, but the alignment story remains synthetic and should not be treated as a real bank portfolio result.
- The SDA section still emits a many-to-many join warning because multiple matched loan rows map to the same sector-year emission series; the output is usable for this demo but should be cleaned for production work.
- Second-best alternative if the full pipeline had stayed blocked would have been to freeze market-share results only and postpone cement and steel, but that would have weakened the later TRISK bridge.

07 / ERRORS / WARNINGS / FLAGS

Errors / Warnings / Flags Encountered

- Earlier progress notes identified a completion blocker in the Vietnam pipeline; rerunning on the current repo state showed the script now completes end to end.
- Two r2dii.plot messages note that technology-share charts default to plotting extreme years only; these are informational, not failures.

- The SDA step still warns about an expected many-to-many join between matched loans and sector-year emission-factor rows.

08 / OPEN QUESTIONS / NEXT PHASE SEEDS

Open Questions / Next Phase Seeds

- Should the SDA join be reworked into an explicit borrower-to-parent mapping table before any production use?
- Should the next iteration freeze an immutable PACTA output snapshot for TRISK runs to avoid accidental drift?
- How much of the existing Vietnam report should be merged into the combined PACTA plus TRISK client artifact versus kept separate?